

| Question | Answer                        | Mark | Mark scheme                                                                                                                                                                                                                                                                                        | Additional guidance                     |
|----------|-------------------------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| 24       | $\frac{5x^2-23x+50}{9x^2-3x}$ | P1   | for using $y = 3x$ or $x = \frac{1}{3}y$ to obtain an expression for a probability in one variable                                                                                                                                                                                                 | This may be awarded at any time         |
|          |                               | P1   | for a correct second probability<br>eg $\frac{x-1}{y-1}$ or $\frac{x-1}{3x-1}$ or $\frac{4}{y-1}$ or $\frac{4}{3x-1}$ or $\frac{y-x-6}{y-1}$ or $\frac{2x-6}{3x-1}$                                                                                                                                |                                         |
|          |                               | P1   | for forming a correct product<br>eg $\frac{x}{y} \times \frac{x-1}{y-1}$ or $\frac{x}{3x} \times \frac{x-1}{3x-1}$<br>or $\frac{5}{y} \times \frac{4}{y-1}$ or $\frac{5}{3x} \times \frac{4}{3x-1}$<br>or $\frac{y-x-5}{y} \times \frac{y-x-6}{y-1}$ or $\frac{2x-5}{3x} \times \frac{2x-6}{3x-1}$ |                                         |
|          |                               | P1   | for adding the 3 correct probabilities<br>eg $\frac{x}{y} \times \frac{x-1}{y-1} + \frac{5}{y} \times \frac{4}{y-1} + \frac{y-x-5}{y} \times \frac{y-x-6}{y-1}$<br>or<br>$\frac{x}{3x} \times \frac{x-1}{3x-1} + \frac{5}{3x} \times \frac{4}{3x-1} + \frac{2x-5}{3x} \times \frac{2x-6}{3x-1}$    |                                         |
|          |                               | A1   | $\frac{5x^2-23x+50}{9x^2-3x}$                                                                                                                                                                                                                                                                      | Can be seen after processing of algebra |